MIL–I–5418B
8 JANUARY 1963
SUPERSEDING
MIL–I–5418A
18 AUGUST 1950

MILITARY SPECIFICATION

INDICATOR: AIRSPEED, PITOT STATIC, SENSITIVE, 50–650 KNOTS

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers certain design and all performance requirements for the procurement of one type of sensitive pitot static airspeed indicator, range 50 to 650 knots

1.2 **Classification.** The airspeed indicators shall be of the following types:

Type	*MS Part No*	*Dial Markings*
1	MS 28047–1	FLUORESCENT
2	MS 28047–2	LUSTERLESS WHITE

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of the invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ–P–416 —Plating, Cadmium (Electrodeposited)
DD–G–451 —Glass, Flat and Corrugated, for Glazing Mirrors and other Uses

MILITARY

MIL–P–116 —Preservation, Method of
MIL–N–3336 —Nut, Self Locking, Instrument Mounting
MIL–E–5272 —Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL–C–5541 —Chemical Films for Aluminum and Aluminum Alloys
MIL–S–7742 —Screw Threads, Standard, Optimum Selected series: General Specification for.
MIL–P–7936 —Parts and Equipment, Aeronautical, Preparation for Delivery
MIL–A–8625 —Anodic Coatings, for Aluminum and Aluminum Alloys
MIL–D–70327 —Drawings, Engineering and Associated Lists

STANDARDS

FEDERAL

FED STD No. 595 —Colors

MILITARY

MIL–STD–130 —Identification Marking of U.S. Military Property
MIL–STD–143 —Specifications and Standards, Order of Precedence for the Selection of

FSC 6610

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MILITARY (continued)

MS28047	—Indicator: Airspeed, Pitot Static, Sensitive, 50-650 Knots
MS28105	—Window, Dial—Aircraft Instrument Cover, Glass
MS33558	—Numerals and Letters, Aircraft Instrument Dial, Standard Form of
MS33585	—Pointers, Dial, Standard Design of Aircraft Instrument
MS33586	—Metals, Definition of Dissimilar

PUBLICATIONS

Air Force-Navy No 418	—Aeronautical Bulletin —Differential Pressures

When requesting any of the applicable documents, refer to both title and number. All requests should be made via the cognizant Government inspector. Copies of this specification and other unclassified specifications and drawings required by contractors in connection with specific procurement functions should be obtained upon application to the Commanding Officer, Naval Supply Depot (Code DCS), 5801 Tabor Avenue, Philadelphia 20, Pennsylvania All other documents should be obtained from the procuring activity or as directed by the contracting officer)

3. REQUIREMENTS

3.1 Qualification. The airspeed indicator furnished under this specification shall be a product which has been tested, and has passed the qualification tests specified herein and has been listed on, or approved for listing on, the applicable qualified products list

3.2 Selection of government documents. Specifications and Standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143 except as provided in 3 2.1 and 3 2 2

3.2.1 *Commercial parts.* Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts In any case, commercial utility parts, like screws, bolts, nuts, cotter pins, having suitable properties, may be used provided:

a. They can be replaced by the standard parts (MS or AN) without alteration.

b. The corresponding standard part numbers are referenced in the parts list and if practicable, on the contractor's drawings

3.2.2 *Standard parts.* With the exception of 3 2 1, MS and AN standard parts shall be used where they suit the purpose They shall be identified on the drawings by their part numbers

3.3 Material. Materials shall conform to applicable specifications and shall be as specified herein. Materials for which there are no applicable specifications, or which are not specifically described herein, shall be of the best quality, of the lightest practicable weight and suitable for the purpose intended.

3.3.1 *Critical materials.* Noncritical materials shall be used where practicable Where the use of a critical material is essential to meet specification requirements, the material used shall be the least critical of those which are adequate for the purpose.

3.3.2 *Metals.* Metals shall be of the corrosion-resistant type, or shall be suitably protected as specified herein to resist corrosion due to fuels, salt spray or atmospheric conditions to which the instrument may be subjected when in storage or during normal service life

3.3.2.1 *Dissimilar metals* Dissimilar metals as defined in MS33586 shall not be used in intimate contact with each other, unless suitably protected against electrolytic corrosion by means of protective coatings or hermetic sealing

3.3.3 *Non-magnetic materials.* Non-magnetic materials shall be used for all parts of the instrument except where magnetic materials are essential.

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3.3.4 *Fungus-Proof materials.* Materials which are nutrients for fungi shall not be used where it is practicable to avoid them Where used they shall be treated with a fungicidal agent acceptable to the procuring activity.

3.3.5 *Corrosive fumes* Materials used in the interior construction of the instrument shall not produce corrosive or deleterious fumes

3.4 **Design and construction.** The airspeed indicator shall be designed to satisfactorily operate in the range of 50-650 knots. It shall be so constructed that no parts will work loose in service and will withstand the normal strains, jars, vibrations and such other conditions as are incident to shipping, storage, installation and service without failure

3.4.1 *Maintenance* The design shall be reasonably simple to facilitate as much as possible disassembly, repair or overhaul, service maintenance, and reassembly by those tools and items of maintenance equipment which are normally available as commercial standards without requiring special tools and fixtures

3.4.2 *Case.* The outline dimensions of the indicator case shall conform to MS28047 The case shall be made of light weight, non-magnetic metal, uniform in texture, have a smooth surface, and be pressure tight The pressure connection of the case shall be centered and the static connection shall be off-center The case shall be finished with a lusterless, black finish of Color No 37038 of FED STD No 595 The finish material shall be of a durable type to withstand usage encountered in service

3.4.2.1 *Insert reinforcement.* The inside of the case around each insert shall be strengthened by a reinforcement to avoid damage to the case when the connection to either insert is tightened during the installation of the indicator

3.4.2.2 *Application of pressure* The pitot pressure shall be applied to the inside of the diaphragm capsule and the static pressure to the interior of the case.

3.4.2.3 *Capillary tubing* A sufficient length of approved capillary tubing having a minimum inside diameter of .020 inch, or some other acceptable restriction, shall be incorporated in the instrument case between the pitot connection in the rear of the case and the diaphragm capsule The effect of this restriction shall be such that the indicator will meet the requirements of the "Damping tests" specified herein.

3.4.2.4 *Standard pressure.* The standard pressure used in calibrating airspeed indicators for use with pitot-static tubes, shall be in accordance with the values specified in ANA Bulletin No. 418

3.4.2.5 *Plug.* Each static and pitot pressure connection shall be provided with a vented plug of material and a design acceptable to the procuring activity The venthole shall be 1/16±1/64 inch in diameter.

3.4.2.6 *Case connection markings.* The back of the case adjacent to the applicable connection shall be marked with the letter "P" for the pressure connection and with the word "STATIC" printed twice, forming a ring around the static connection The words "CAUTION: DO NOT BLOW IN TUBES" shall also be marked on the back of the case. All the markings shall be durable to withstand the usage encountered in service The markings shall be white, Color 37875 of FED STD No 595

3.4.3 *Dial.*

3.4.3.1 *Dial plate* The dial plate shall be made of non-magnetic metal It shall be securely fastened into the case or upon the frame of the mechanism in such a manner that it will not loosen or turn when the indicator is vibrated At least two screws shall be used to attach the dial plate, so located that they will not interfere with any dial marking or prevent locating any dial marking in the proper location.

3.4.3.2 *Window.* The window in the main dial shall be large enough to keep the hundreds numeral visible as long as the instrument is indicating in that range

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3.4.3.3 *Visibility of dial.* The pointer, numerals, at least 1/16 inch of the shortest graduations and all other specified markings on the dial shall be visible from any point within the frustum of a cone whose side makes an angle of 30° with a perpendicular to the dial and whose small diameter is the aperture of the case

3.4.3.4 *Cover glass.* The cover glass shall be flat and its quality shall be in accordance with Specification DD-G-451, Type II, Quality AA. The glass shall not be fluorescent when exposed to ultraviolet light or rays in the 365 millimicrons region. The dimensions shall be in accordance with MS28105-16.

3.4.3.4.1 *Cover glass mounting.* The cover glass shall be properly sealed and means shall be provided both for its replacement and to prevent fogging or the accumulation of moisture on the inside surface under extremes of atmospheric conditions. The distance between the front surface of the mounting flange and the outside surface of the cover glass shall not exceed 0.062 inch. The distance between the dial plate and the inside surface of the cover glass shall be the minimum practicable and shall not exceed 0.125 inch.

3.4.3.5 *Dial markings.* The dimensions of the dial markings and the color of the marking material used shall be as shown in Figure 1. The style and proportions of the letters and numerals placed on the dial shall conform to MS33558 The lusterless white material shall be of Color 37875 of FED STD No 595 The lusterless black material shall be of Color 37038 of FED STD No. 595. The dial painting material shall be of a quality sufficient to insure durability of the markings in service use.

3.4.3.5.1 *Numerals.* Numerals shall distinctly indicate the graduation to which each applies If practicable, each numeral shall be placed so that the center of mass of the numeral is on the radial line joining the appropriate graduation and the center of the dial. Any confusion resulting in doubt as to which graduation the numeral applies shall be cause for rejection. When several numerals are used in one group, the space between the numerals shall be not less than 1/64th inch.

3.4.4 *Pointer.* The pointer shall conform to MS33585-3 The shaded portion of the pointer shall be finished with lusterless white material of color 37875 of FED STD No. 595. The unshaded portion shall be finished with lusterless black material of color 37038 of FED STD No 595

3.4.4.1 *Construction.* The pointer shall be as light as practicable and sufficiently rigid to prevent oscillation under vibration It shall be firmly attached to the mechanism, and shall be readily adjustable.

3.4.4.2 *Limitation of pointer movement.* The pointer movement shall be limited by suitable stops in the mechanism in such a way that the pointer will not rotate counterclockwise beyond the 45-knot graduation on the dial. The design of the stops shall be subject to the approval of the procuring activity.

3.4.5 *Weight* The indicator shall be so designed that when completely assembled the weight shall not exceed 1.25 pounds.

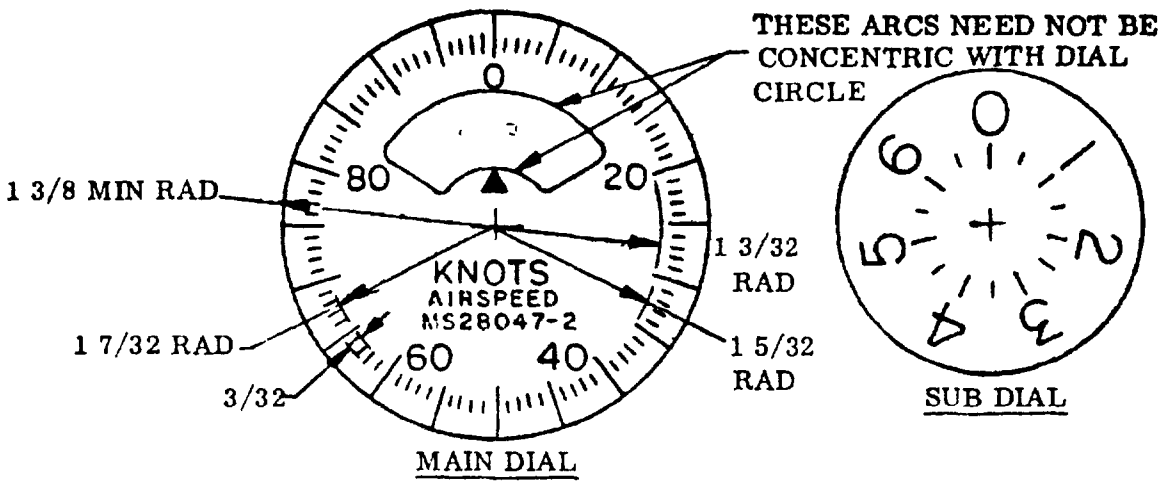
3.5 **Performance.** The airspeed indicators shall perform satisfactorily when subjected to the tests specified in Section 4 of this specification.

3.6 **Physical requirements.**

3.6.1 *Protective treatment.* When materials are used in the construction of the indicator that are subject to corrosion in salt air or other atmospheric conditions likely to occur during service usage, they shall be protected against such corrosion in a manner that will in no way prevent conformance with the requirements of this specification. The use of any protective coating that will crack, chip or scale during normal service life or due to extremes of atmospheric conditions shall be avoided.

3.6.2 *Aluminum alloy parts.* Where practicable, aluminum alloy parts shall be covered with an anodic film conforming to Specification MIL-A-8625 except that the dial, if of aluminum alloy, small holes, pipe

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ALL GRADUATIONS ARE PARTIAL RADIAL LINES.
DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCE· FRACTIONS ±1/64.

MARKING	HEIGHT OR LENGTH ± .010	WIDTH OF LINE OR GRADUATION ± .005	FINISH COLOR
MAIN DIAL			
Numerals 0, 20, 40, 60 80	187	031	
Graduations corresponding to 0 10 20, 30 40 50, 60 70, 80 90 knot readings	As shown on figure	031	
Graduations corresponding to 5, 15 25, 35 45 55, 65 75, 85, 95 knot readings	As shown on figure	016	
One knot graduations from 1 to 99 knots inclusive	As shown on figure	016	
Triangular index	156	156	LUSTERLESS WHITE or FLUORESCENT
SUB DIAL			
Numerals 0, 1, 2, 3, 4, 5, 6	.312 (min)	031	
Graduations corresponding to numerals 0, 1, 2, 3, 4, 5, 6	.125 (min) Visible thru Main Dial Cutout	031	
Graduations between numerals 0 & 1, 1 & 2, 2 & 3, 3 & 4, 4 & 5, 5 & 6, 6 & 0	063 (min) Visible thru Main Dial Cutout	.016	
Lettering "Knots"	188	016	
Lettering "MS28047-2"	047	016	LUSTERLESS BLACK
Lettering "Airspeed"	.047	.016	
Background of dial	— —	— —	

FIGURE 1—*Dial Marking*

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threads and case inserts need not be anodized. Aluminum alloys which do not anodize satisfactorily shall be coated with chemical film in accordance with Specification MIL–C–5541.

3.6.3 *Iron and steel parts.* Where practicable, iron and steel parts shall be cadmium plated in accordance with Specification QQ–P 416 Parts in a confined space in the presence of organic material shall be tin-cadmium coated

3.6.4 *Screw threads* Screw threads 0 060 inch or larger in diameter shall be in accordance with Specification MIL–S–7742

3.6.5 *Gears* Gear assemblies shall be properly aligned and meshed and shall be operable without interference, tight spots, loose spots or other irregularities Where required for accurate adjustments, gear assemblies shall be free from backlash

3.6.6 *Dimensions* Dimensions and tolerances not specified, shall be as close as is consistent with the best shop practices Where dimensions and tolerances may effect the interchangeability, operation or performance of the instrument, they shall be held or limited accordingly.

3.6.7 *Workmanship* The instrument, shall be constructed and finished to produce an instrument free from all defects which would affect proper functioning in service Particular attention shall be given to neatness and thoroughness of soldering, marking of parts and assemblies, welding and brazing, painting, riveting, machine-screw assemblies and freedom of parts from burrs and sharp edges.

3.6.8 *Cleaning* The instrument shall be thoroughly cleaned of loose, spattered or excess solder, resin flash that may crumble, metal chips and other foreign matter, after final assembly

3 7 **Identification of the product.**

3.7.1 *Nameplate.* A nameplate shall be securely attached to the exterior of the case and shall be durably marked to withstand usage encountered in service The nameplate shall be legibly marked in accordance with MIL–STD–130.

3.7.2 *Manufacturer's part number* The manufacturer's part number on the nameplate shall be identical with the manufacturer's engineering production drawing number including applicable dash numbers if the drawing is tabulated and covers more than one part.

3.7.3 *Interchangeability.* All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance Changes in manufacturer's part number shall be governed by the drawing number requirements of Specification MIL–D–70327.

3.7.4 *Use of military designations* Military designations shall not be applied to a product, except for Qualification Test Samples nor referred to in correspondence or sales matter, until notification has been received from the activity responsible for qualification

3.8 **Installation instructions.** The contractor shall furnish with each instrument one copy of the installation instructions with illustrations and diagrams Prior to printing, two copies shall be furnished to the procuring activity for approval Whenever possible the instructions shall be arranged to require only one sheet of paper either 8½ by 11 or 11 by 17 inches.

3.8.1 *Removable spring nuts* Removable spring nuts of a type conforming to Specification MIL–N–3336, shall be furnished in sufficient quality to fit the mounting lugs of the instrument.

3.8.2 *Envelope* An envelope furnished by the contractor, containing installation instructions and spring nuts shall be packaged with each instrument and shall be marked with the following information

IMPORTANT
THIS ENVELOPE CONTAINS
INSTALLATION INSTRUCTIONS
AND SPRING NUTS

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4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification when such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of tests. The Inspection and Testing of the airspeed indicators shall be classified as follows:

a. *Qualification tests:* Qualification tests are those tests performed on sample instruments submitted for qualification as a satisfactory product.

b *Acceptance tests* Acceptance tests are those tests performed on instruments manufactured and submitted for acceptance under contract

4.3 *Qualification tests* The qualification tests of the instrument shall consist of the tests listed below, conducted in the order given.

Individual Tests	Fungus Resistance Test
Sampling Plan A Tests	Seasoning Test
Sampling Plan B Tests	Salt Spray Test
Acceleration Test	Aging Test
Humidity Test	Mounting Lugs Test

4.3.1 *Qualification test instructions* Qualification test samples shall consist of three indicators, of each manufacturer's part number to be furnished in accordance with this specification. The instruments submitted for testing shall not have been previously tested except for the Individual Tests The samples shall be forwarded to the testing laboratory designated in the letter of authorization

4.3.2 *Test sample identification.* The test samples shall be plainly identified by securely attached, durable tags marked with the information listed below.

Sample for Qualification Testing
Pitot Static Sensitive Airspeed Indicator (50-650 Knots)
Submitted by *(Manufacturer's name, date)* for Qualification Testing in accordance with Specification MIL–I–5418 under authorization (reference letter authorizing tests)
MS28047– (Add proper dash number)
Manufacturer's Part Number

4.4 Acceptance tests. The acceptance tests of the instruments shall consist of the Individual Tests and the Sampling Plans of this specification. The contractor shall furnish all samples and shall be responsible for accomplishing the required tests. When inspection is conducted at the contractor's plant, all inspection and testing shall be under the supervision of the Government Inspector The contractor shall furnish test reports showing quantitative results for all tests required by this specification, signed by an authorized representative of the contractor or laboratory, as applicable Acceptance or approval of material during the course of manufacture shall in no case be construed as a guarantee of the acceptance of the finished product

4.4.1 *Individual tests* The individual tests shall consist of the tests listed below, conducted in the order given, on each instrument.

Examination of the Product
Leakage Test
Scale Error at Room Temperature Test
Friction Error Test
Position Error Test

4.4.2 *Sampling plans and tests.*

4.4.2.1 *Sampling plan A.* Samples shall consist of three airspeed indicators selected at random by the Inspector from each lot of 100 or less which have passed the Individual Tests. A lot shall consist of identical in-

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struments with the same manufacturer's part number, manufactured under substantially the same conditions and submitted at substantially the same time. Instruments which have been subjected to the Sampling Plan A tests shall not be delivered on contract until they have been reconditioned and resubmitted to all Individual Tests

4.4.2.1.1 *Sampling plan A tests* The sampling plan A tests shall consist of the tests listed below, conducted in the order given, on each sample

Damping Test
Weight Test
Magnetic Effect Test
Pointer Zero Position Error Test
Low Temperature Exposure Test
High Temperature Exposure Test

4.4.2.2 *Sampling plan B* Unless otherwise specified, the samples shall consist of three indicators selected at random by the Inspector from the first 15 instruments produced on contract within ten days after manufacture The samples shall be forwarded to the laboratory specified herein (see 6.2) and shall be plainly identified by securely attached, durable tags marked with the following information:

Pitot Static Sensitive Airspeed Indicator (50-650 Knots)

Submitted by *(manufacturer's name, date)* for production acceptance testing in accordance with Contract/Order No ——.

4.4.2.2.1 *Sampling plan B tests.* The sampling plan B tests shall consist of the tests listed below, conducted in the order given, on each sample.

Vibration Test
Overpressure Test

4.4.3 *Rejection and retest* The instructions on reworking, refurbishing or resubmitting rejected items shall be as specified in the General Provisions of the contract order.

4.5 Test conditions

4.5.1 *Standard atmospheric conditions.* Whenever the pressure and temperature existing at the time of test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches of mercury) and at room temperature (approximately 25° C (77° F)). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

4.5.2 *Tapping.* Unless otherwise specified, the indicator shall be gently tapped or vibrated before a test reading is taken.

4.5.3 *Attitude* Unless otherwise specified, the indicator shall be tested in normal operating position.

4.5.4 *Vibration.* A vibration stand shall be used which will vibrate at any desired frequency between 500 and 3,000 cycles per minute and shall subject the indicator to such vibration that a point on the vibration stand will describe, in a plane inclined 45° to the horizontal plane, a circle of the diameter specified herein

4.5.5 *Test values.* The test points, allowable error and pressure calibration values in inches of water and mercury shall be as shown in Tables I and II

4.6 Test methods

4.6.1 *Examination of the product.* Each indicator shall be examined externally to determine conformance with the applicable drawings and with all the requirements of this specification not covered by tests

4.6.2 *Leakage*

4.6.2.1 *Case leak* The pitot and static pressure connections of the indicator shall be joined with a "Y" connection and shall be connected together to the same mercury manometer and sources of suction and pressure A suction of 15 inches of mercury shall be applied to the pitot and static pressure connections of the indicator. With the source disconnected for a period of 60 seconds, the difference between the mercury levels in the manometer shall not change by more than 0 1 of an inch. This procedure shall be re-

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TABLE I

Test Values

Tests	Tolerances (Values in knots unless otherwise specified)
DAMPING Time required from full scale deflection to 100 knots	1.5 ±0.2 sec.
LOW TEMPERATURE EXPOSURE AND OPERATION	2.5
HIGH TEMPERATURE OPERATION	2.5
AGING	2.5
VIBRATION FAILURE Test differential pressure equivalent to 320 knots	2.5
FRICTION	3.5
POSITION ERROR	2.5
POINTER ZERO POSITION Indication at 45 knots	+5, -0
VIBRATION ERROR Test differential pressure equivalent to 120, 320, and 520 knots	2.5
OVERPRESSURE Pitot pressure connection; 30 inches of mercury. Static pressure connection; 6.0 inches of water	2.5
SEASONING	2.5

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TABLE II

Scale Errors at Room Temperature

Speed	Pressure in Inch.		Allowable Scale errors	Speed	Pressure in Inch.	Allowable Scale errors
Knots	H_2O	Hg	Knots	Knots	Hg	Knots
50*	1.634		4.0	360	6.689	4.0
60	2.354		2.0	370	7.095	4.0
70	3.207		2.0	380	7.516	4.0
80*	4.192		2.0	390	7.951	4.0
90	5.310		2.0	400*	8.401	4.0
100	6 563		2.0	410	8.866	4.0
110	7 951		2.0	420	9.348	4.0
120*	9 475		2.0	430	9.845	4.0
130	11.14		2.0	440	10.36	4.0
140	12.94		2.5	450*	10.89	4.0
150*	14.87		2.5	460	11.44	4.0
160	16.95		2.5	470	12.00	4.0
170	19.17		2 5	480	12.58	4.0
180*	21 54		2 5	490	13.18	4.0
190	24.05		2.5	500*	13.80	4.0
200		1 962	3 0	510	14.44	5.0
210		2.169	3.0	520	15.10	5.0
220		2.356	3.0	530	15.78	5.0
230		2.614	3.0	540	16.47	5.0
240		2.854	3 0	550*	17.19	5.0
250*		3.106	3.0	560	17.93	5.0
260		3.369	4.0	570	18.70	5.0
270		3.644	4.0	580	19.48	5.0
280		3.931	4.0	590	20.29	5.0
290		4.231	4.0	600*	21.12	5.0
300*		4.543	4.0	610	21.97	5.0
310		4.867	4.0	620	22.85	5.0
320		5.205	4.0	630	23.76	5.0
330		5.556	4 0	640	24.69	5.0
340		5.920	4.0	650*	25.64	5.0
350*		6.298	4.0			

* Test points

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peated with a pressure of 10 inches of mercury. During a period of 60 seconds, the difference in the mercury levels in the manometer shall not change by more than 0 1 of an inch

4.6.2.2 *Diaphragm capsule leak.* With the static connection open to the atmosphere, a pressure sufficient to produce approximately full scale deflection of the pointer, shall be applied to the pitot connection of the indicator at which point the connecting tubing shall be pinched off or otherwise completely sealed During a period of 1 minute, the pointer shall not change its position by more than 1 knot

4 6.3 *Scale error at room temperature* (See Section 6). The indicator shall be tested for scale errors at the points of the scale indicated by asterisks in table II The test shall be made by subjecting the indicator to the pressures specified to produce these readings, first with pressures increasing, then with pressures decreasing. With pressures increasing, the pressure shall be brought up to, but shall not exceed the pressures specified to give the desired reading, and with pressures decreasing the pressure shall be brought down to, but shall not fall below the pressures specified to give the desired reading The errors at the test points shall not exceed those specified in table II When desired by the Inspector, the indicator shall be tested for scale error at any additional point or points on the scale The errors at these additional test points shall not exceed those specified in table II

4.6.4 *Friction error.* The indicator shall be tested for friction at every other one of the points indicated by an asterisk in table II, beginning with the first asterisk The pressure shall be increased to bring the pointer approximately to the desired reading, and then held constant while two readings are taken, the first before the indicator is tapped, the second after. The difference between any two such readings shall not exceed the values specified in table I. The pointer shall move smoothly while the pressure is varied uniformly without vibration of the indicator This test may be combined with the test for scale error at room temperature

4.6.5 *Position error.* The indicator shall be tested as specified in the Friction Test except that at each test point, it shall be held in the following positions:

a Rotated from level position through 45 degrees to the rear (dial tilted 45 degrees backward from vertical position)
b. Rotated from level position through 45 degrees to the front (dial tilted 45 degrees forward from vertical position).
c. Rotated from level position around major axis through 45 degrees to the right (dial vertical)
d. Rotated from level position around major axis through 45 degrees to the left (dial vertical).

The error at each position shall not exceed the tolerance specified in Table I.

4.6.6 *Damping.* Sufficient pressure shall be applied to the pitot connection to produce full scale deflection of the airspeed pointer The pressure shall suddenly be released. The time required for the airspeed pointer to reach the value specified in table I shall be equal to the time and within the tolerance specified therein

4.6.7 *Weight* The indicator shall be weighed completely assembled and without connecting fittings and mounting screws The weight of the indicator shall not exceed the value specified in paragraph 3 4 5

4.6.8 *Magnetic effect.* The indicator shall be revolved about a short bar magnet compass with the nearest part 5½ inches from the bar magnet, and in a vertical plane which is a perpendicular bisector of the North-South axis of the bar magnet Starting directly under the compass, the indicator shall be held in positions 0, 45, 90, 135, 180, 225, 270, and 315 degrees from the initial position At each of these positions the indicator shall be rotated on its horizontal axis until it is in its normal operating position. The horizontal magnetic field intensity shall be

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0 17 to 0 19 oersted. The deflection of the compass at any of the specified positions shall not exceed one degree

4.6.9 *Pointer zero position error.* The zero position of the pointer with no differential pressure applied to the indicator shall be within the values specified in table I.

4.6.10 *Low temperature exposure.* The indicator shall be tested in accordance with Procedure II of Specification MIL–E–5272 except that after the 24-hour period and while still at the low temperature, the indicator shall be tested for scale error as specified in the Scale Error test The errors at the test points shall not exceed those specified in table II by more than the values specified in table I After the indicator has returned to room temperature, it shall meet all the requirements specified in the Individual tests

4.6.11 *High temperature exposure.* The indicator shall be tested in accordance with Procedure II of Specification MIL–E–5272. After the indicator has returned to room temperature, it shall meet all the requirements specified in Individual tests

4.6.12 *Vibration* The indicator shall be tested in accordance with Procedure IV of Specification MIL–E–5272 This test shall be conducted while the differential pressures specified in table I are applied to the indicator While the indicator is being vibrated, the maximum total spread of pointer vibration and the pointer variation from its original position shall not exceed the values specified in table I

4.6.13 *Overpressure*

4.6.13.1 The pressure specified in table I shall be applied to the pitot pressure connection of the indicator for a period of 10 minutes

4.6 13.2 The pressure specified in table I shall be applied to the static connection of the indicator for a period of 10 minutes

4.6.13.3 Not less than 5 minutes following the application of Overpressure tests, the indicator shall be tested for scale error as specified in the Scale Error at Room Temperature test. The errors at the test points shall not exceed those specified in table II by more than the value specified in table I.

4.6.14 *Acceleration.* The indicator shall be mounted in its normal operating position on a rotating arm, with the rear of the case toward the center of rotation, and shall be subjected to an acceleration of 20g's for a period of 30 seconds. This test shall be repeated with the indicator mounted successively in the following two positions: (1) rotated 90 degrees in azimuth with respect to the original position and (2) dial horizontal, with the zero graduation toward the center of rotation At the end of this acceleration, the indicator shall meet all the requirements specified in the Individual tests. No damage to the indicator shall result from this test.

4.6.15 *Humidity.* The indicator shall be mounted in its normal operating position in a chamber and maintained at a temperature of 65° C (149° F) and a minimum relative humidity of 95 percent for a period of 6 hours External connections shall be made to the indicator to simulate installed conditions. After this period, the heat shall be shut off and the indicator shall be allowed to cool for a period of 18 hours in this atmosphere, in which the humidity rises to 100 percent as the temperature decreases. Precaution shall be taken to prevent dripping on the indicator. The complete procedure shall be repeated 10 times. There shall be no evidence of corrosion of the internal or external parts of the indicator which would affect the performance of the instrument

4.6.16 *Fungus resistance.* The indicator shall be tested in accordance with Procedure I of Specification MIL–E–5272, except that the test period shall be 14 days. At the option of the qualifying agency, tests of component parts of the item may be accepted in lieu of, or in addition to, tests of the assembled instrument. After this test, the indicator shall be subjected to and shall meet the requirements specified for the Aging test. There shall be no deterioration which would adversely affect subsequent operation This test shall be run simultaneously with the Aging test.

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4.6.17 *Seasoning.* The indicator shall be subjected to 1,000 applications of a differential pressure sufficient to produce approximately full scale deflection. These applications shall be made at a rate not greater than 6 per minute. Not less than 48 hours following this test, the indicator shall be tested for scale errors as specified in the Scale Error at Room Temperature test. The errors at the test points shall not exceed those specified in table II by more than the values specified in table I.

4.6.18. *Salt spray.* The indicator shall be tested in accordance with Procedure I of Specification MIL–E–5272 for a period of 50 hours. External connections shall be made to the indicator to simulate installed conditions At the end of the 50-hour period, no damage to any part of the indicator which would affect the performance of the instrument shall result from this test The indicator shall then be subjected to and shall meet all the requirements specified in the Individual tests.

4.6.19 *Aging.* The indicator shall be retained for a period of not less than 30 days after having been subjected to the Seasoning test, and then tested for Scale Error at Room Temperature. The scale errors in this test shall not differ from those obtained 48 hours after the 1,000 applications of differential pressure specified in the Seasoning test by more than the value specified in table I.

4.6.20 *Mounting lugs* The case of the indicator shall be mounted face downward on the movable head of a suitable testing machine with the face of the case in a horizontal plane so that the mounting lugs receive no added support A suitable pin shall be inserted through the hole in the mounting lug and attached to a pull strap in the stationary head of the machine A load of 175 pounds shall be applied for one minute on each lug in a direction toward the front of the case. The lugs shall withstand the applied load without fracture and there shall be no damage to any part of the instrument.

5. PREPARATION FOR DELIVERY

5.1 *Packaging*

5.1.1 *Level A.* The indicator shall be packaged in accordance with Specification MIL–P–7936 The method of preservation shall be in accordance with Specification MIL–P–116, Method IId omitting preservative compound and using metal reusable containers

5.1.2 *Level C.* When this level is required, packaging shall be in accordance with standard commercial practice

5.2 *Packing*

5.2.1 *Levels A and B.* The instrument shall be packed in accordance with Specification MIL–P–7936 The level or levels required shall be as specified in the invitation for bid or contract

5.2.2 *Level C* The instrument shall be packed to insure that the shipment arrives in a satisfactory condition at destination. The shipment shall conform to the applicable carriers' rules and regulations in effect at the time of shipment.

5.3 *Marking.* The interior and exterior containers shall be marked as specified in Specification MIL–P–7936.

5.3.1 *Precautionary marking* The following precautionary marking shall appear on two opposite sides of each interior package and shipping container whenever practicable depending on the size of the carton:

FRAGILE

DELICATE INSTRUMENTS

HANDLE WITH CARE

6. NOTES

6.1 Intended use. The airspeed indicators covered by this specification are intended for use by flight engineers on aircraft to indicate the speed relative to the air at sea level and, when at any other altitude, the equivalent sea level speed corresponding to the actual sustaining force of the air.

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, Number, and Date of this Specification.
b. The quantity and MS part number of the instrument desired.
c. Whether Sampling Plan B is to be omitted and if not, add the name and address of the laboratory to which the samples should be submitted
d. Levels of packaging (see 5.1 and packing (see 5.2) desired

6.3 *Drawing requirements* The attention of contractors is invited to the fact that Government contracts for equipment, whether procured directly by the Government or through another contractor, require that all drawings submitted must be in accordance with the requirements of specification MIL–D–70327

6.4 *Definitions*

6.4.1 *Scale error.* By "scale error" is meant the difference between the indication of the instrument and the calibrated airspeed corresponding to the applied differential pressure as specified in ANA Bulletin No 418

6.4.2 *External connection* When external connections to simulate installed conditions are called for, they shall consist of ten feet of ¼-inch copper or aluminum tubing so coiled that moisture can drain out the open end Similar coils shall be connected to the pitot and static pressure connection.

6.5 *Qualification.* With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date The attention of the suppliers is called to this requirement and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Bureau of Naval Weapons, Department of the Navy, Washington 25, D. C , and information pertaining to qualification of products may be obtained from that activity

6.5.1 It is to be understood that upon receipt of the Letter of Authorization, samples shall be furnished at no cost to the Government, and that the manufacturer shall pay the transportation charges to and from the designated point where the tests are to be made In the case of failure of the sample or samples submitted, consideration will be given to the request of the manufacturer for additional tests only after it has been clearly shown that changes have been made in the product which the Government considers sufficient to warrant additional tests

6.5.2 It is to be understood that indicators supplied under contract shall be identical in every respect to the sample tested and found satisfactory, except for changes previously approved by the Government. Any unapproved changes from the qualification sample shall constitute cause for rejection

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever, and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodian:
Army—Transportation Corps
Navy—Bureau of Naval Weapons

Preparing activity:
Navy—Bureau of Naval Weapons

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